

DIGITAL HEALTH DRIVEN PREVENTION AS A CLINICAL REALITY

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Summary: The increasing burden that non-communicable diseases (NCDs) inflict on health care systems has continued to rise in recent decades. As many NCDs are preventable, early identification and intervention is key. Progress in the fields of data science and analytics, including recent utilisation of artificial intelligence (AI) in health care, have been identified as game changers in making preventive endeavours effective. Yet, the use of digital health at scale in practice is limited. Based on the experience of an integrated health care payer-provider system in Israel, several key attributes to successful prediction-based preventive proactive interventions have been identified.

Keywords: Prevention, Prediction, Causal inference, Digital Health, Israel

Introduction

NCDs are a leading cause of mortality and place a growing burden on the health care system, with global incidence and prevalence increasing.¹ With a rise in the numbers of patients who are ageing and have multi-morbidities, health systems are facing growing demands, which available resources are unable to meet. Effective prevention measures can reduce this burden and extend healthy life expectancy through the execution of proactive interventions, but realising this paradigm at a population level is challenging both financially and logistically.²

The increasing availability of longitudinal digital health care data, together with advancements in data science and most notably in AI, create new opportunities

to analyse, augment, and alter health care delivery to provide more efficient preventive and proactive medicine.³ As a by-product in the past few years, there has been a surge in new digital health tools for diagnosis, prognosis, therapeutics, care management, and care delivery.⁴

Yet, despite a plethora of readily available digital health tools, including hundreds of thousands of mobile health applications, with hundreds more launched on a daily basis,⁵ successful attempts at harnessing AI to improve health care at scale in clinical practice are rare.^{6,7} Creating predictive models with strong predictive performance, for example, is far from enough to ensure their use and acceptance by health care providers.⁸

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Creating and integrating predictive models in the Israeli health care system

The prediction models in use within health organisations in Israel provide examples from which it is possible to learn about their design, implementation, and ability to scale up. In Israel, all citizens have universal access to health care through one of four payer-provider health care organisations. The largest of these organisations, Clalit Health Services (Clalit), serves over 4.4 million members, which constitute over half of the Israeli population. Clalit benefits from being a fully integrated system consisting of 14 hospitals, over 1,500 primary care clinics, while also operating its own laboratories, pharmacies, imaging services and other facilities. Coupled with an extremely low annual member attrition rate (less than 2%) and capitation-based budget, an embedded economical and operational incentive for long-term preventive care has been created.⁹

“critical steps are overlooked

Data on all Clalit members are recorded in individual electronic health records (EHRs), which date back to the 1990s. All socio-demographic and administrative data, inpatient and outpatient details, health care utilisation, medical history information, and medical encounters are also recorded in the EHRs. The longitudinal data has been extremely influential in creating data-driven tools and a “living lab” in which to implement them.

Data derived from the EHRs, coupled with the work of a multi-disciplinary group of physicians, epidemiologists and data-scientists from the Clalit Research Institute (CRI), has allowed Clalit to create predictive models, and the integration within Clalit made it possible to implant them into actual practice at scale, for over a decade. Such predictive proactive prevention plans aim at tackling and preventing key diseases associated with

the global burden of disease, including chronic kidney disease progression, asthma attacks, onset of type-2 diabetes among pre-diabetics, acute deterioration of geriatric patients, 30-day readmissions and winter morbidity (pneumonia and influenza).¹⁰ This work serves as a basis for CRI's work as the WHO Collaborating Center on Non-Communicable Diseases Research Prevention and Control.

Understanding the key challenges

While prediction tools have great potential to enable proactive and preventive interventions, it is essential to understand their limitations. Our experience suggests that several key factors impact this process, and in too many of these suggested new interventions these critical steps are overlooked.

Workflow engineering and stakeholder involvement

The first step, and often the most critical component, in designing a new prediction-based intervention is workflow engineering to allow the process to be streamlined with the daily work of practising clinicians. It is also critical to involve all stakeholders in ensuring that the intervention addresses a well-defined unmet need for administration as well as the end-user. When designing the readmission prediction model that was implemented in parallel in the ambulatory and inpatient setting, it was an outcome of intense deliberations with clinical staff, including key opinion leaders and practising staff, as well as administrators at all levels of the system. This has proven immensely important in establishing a clear new workflow in each of the settings and allocating resources when it required adjustment.

It is important for administrators to be aware that predictive proactive models of population health eventually get translated into intervention lists, which are essentially a new workload for clinicians. It is of value to address the issue of “impactability” – i.e. the likelihood of the intervention to positively affect the selected individuals, and not only risk. Carving out the likely low-impact sub-groups from top risk tiers reduces the burden of the unduly long lists and reduces

the frustration of physicians facing patients that may be at risk but are unlikely to benefit from any intervention as the deterioration of their chronic condition is intractable or non modifiable.

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Select simple models when possible

Another key challenge in introducing prediction tools into practice is current pushback from the care providers who should use these tools on a daily basis. There is a fine balance between the need to create prediction tools with explainable methodologies, (i.e. why a certain decision was made, namely, “white box models”, such as simple decision trees or logistic regression) and the motivation to utilise the more complex prediction algorithms that can sometimes produce better performance but are more difficult to explain (“black boxes” such as random forests or neural networks, that can use the entire medical history data).

Over the years, Clalit has deployed models of both kinds. The factors that affected the choice of model type usually were due to the clinicians' opinions, as well as relative performance of the different model types. When the differences in performance between simple and complex models are minor, the simpler model is always preferred. It is also critical not to be only data-driven, but to collect key features from the clinical experience of the end-users, and see if those can be reasonably included in the prediction to increase its face validity through the lens of these same end-users.

It is likely that over time, clinicians will learn to trust prediction models, and thus more black box models could be integrated into practice. In the meantime,

simpler models are often good enough, or alternatively technologies that explain the leading features in black boxes models, such as the SHAP (SHapley Additive exPlanations) method, can be used to gain clinicians' trust.

Identify when a different toolset is necessary

Finally, it is important to be very cautious when the challenge to tackle with a predictive model is prescriptive rather than predictive – i.e. when the aim is not to classify patients based on their risk but rather to predict the most favourable impact among several treatment options. The latter requires a different methodological toolset, one which stems from the rapidly developing field of causal inference, and simple predictions may lead to erroneous recommendations. Unlike prediction models, where the performance of the model can be evaluated using test and validation sets, it is far more complicated to determine the performance of a causal inference algorithm when based on observational data. Even when employing multiple statistical techniques to minimise the amount of confounding on the causal effect including minimising differences between the exposed groups, there is always considerable residual confounding.

When properly used, these 'counterfactual assessment' models can serve as strong motivators in changing patient behaviour. CRI is using causal inference in several prescriptive models that are delivered directly to patients in their mobile application, to see how different behavioural changes (e.g. smoking cessation or weight reduction) can alter their life course and reduce their risk of morbidity.

Conclusion

Being able to utilise prediction models more effectively, while also more accurately understand the anticipated impact of different interventions for a specific individual before testing them in practice, requires understanding causal effects. Deducing from massive longitudinal retrospective data such causal effects of where and how it is preferable to intervene by weighing alternative

outcomes will further transform health care in the coming years. These processes will contribute substantially to reaching the next milestone in achieving effective and affordable population health maintenance at scale.

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